Independent Replication: *Quantum Counting* (Brassard, Høyer, Tapp; arXiv:quant-ph/9805082, 1998)

QC-200 Replication Wave, 2026-07-05

July 5, 2026

Abstract

This report is an independent, from-scratch replication of the main algorithmic claim of Brassard, Høyer and Tapp (BHT'98), commonly known as *Quantum Counting*: given a Boolean function $F : \{0, \ldots, N-1\} \to \{0, 1\}$ with $t = |F^{-1}(1)|$ marked items, an estimate $\tilde{t}$ of t can be produced with P oracle queries such that

$$|t - \tilde{t}| \ < \ \frac{2\pi}{P}\sqrt{tN} + \frac{\pi^2}{P^2}N \qquad \text{(Theorem 5 of BHT'98)}$$

with probability at least $8/\pi^2 \approx 0.811$. We implemented the algorithm end-to-end in Qiskit 2.5.0 statevector, ran it on $N = 16$ for $t \in \{1, 2, 4, 8\}$ with $P \in \{16, 32\}$, and observed that (a) the maximum-probability estimate $\tilde{t}$ always rounds to the true t and (b) the empirical success probability under the Theorem's inequality is ≥ 0.87 across all 8 configurations, comfortably exceeding the $8/\pi^2$ analytical floor. **Verdict: REPLICATED.**

1 Paper Summary

The paper contains three algorithmic contributions: (i) a general *amplitude amplification* framework that unifies and generalises Grover's search (Theorem 1–2, quadratic speed-up over the initial success probability a); (ii) an extension of the quadratic speed-up to families of classical heuristics with unknown a (Theorem 3–4); (iii) an *approximate counting / amplitude estimation* algorithm that combines Grover's iteration G_F with the quantum Fourier transform in the style of Shor's phase-estimation (Section 4, algorithm `Count`, Theorem 5–6). Contribution (iii) is the paper's stated main result and the target of this replication.

The `Count`(F, P) algorithm, verbatim from the paper, is:

```
1. |Psi_0> <- W tensor W |0>|0> # Hadamard on both registers
2. |Psi_1> <- C_F |Psi_0> # C_F : |m>|Psi> -> |m>(G_F)^m|Psi>
3. |Psi_2> <- optional measurement of the search register
4. |Psi_3> <- F_P tensor I |Psi_2> # (inverse) QFT on precision register
5. f_tilde <- measure |Psi_3> # if f_tilde > P/2 then f_tilde <- P - f_tilde
6. output N sin^2(f_tilde pi / P)
```

which is exactly *quantum phase estimation of the Grover operator* G_F, whose two non-trivial eigenvalues are $e^{\pm 2i\theta}$ with $\sin^2\theta = t/N$.

1.1 Claims Table

Id	Claim	Type	Tested?
C1	Amplitude amplification quadratic speed-up (Thm. 1–2)	analytical	partial (C1 subsumed by C4)
C2	Quadratic speed-up without knowing a (Thm. 3–4)	analytical	no (out of scope)
C3	`Count`(F, P) is a P-query quantum algorithm for $\tilde{t}$	algorithmic	**yes** (implemented + run)
C4	$\lvert t - \tilde{t}\rvert < \frac{2\pi}{P}\sqrt{tN} + \frac{\pi^2}{P^2}N$ w.p. $\geq 8/\pi^2$ (Thm. 5)	quantitative	**yes** (measured on 8 configs)
C5	Amplitude-estimation variant (Thm. 6)	quantitative	no (Thm. 5 is the direct headline)
C6	Related-error/relative-error corollaries (Cor. 2–3)	quantitative	no (implied by C4)

C3 and C4 are the "one most-checkable number" pair required by the QC-200 brief.

2 Method

2.1 Environment

- Host: `CherryRd` (macOS 15.3.0, Python 3.13).
- Python venv: `./venv` (with `--system-site-packages` so numpy carries over).
- Packages: `qiskit 2.5.0`, `qiskit-aer 0.17.2`, `numpy 2.4.3`.

2.2 Reproducing Commands

```
# from repo root
python3 -m venv venv --system-site-packages
source venv/bin/activate
pip install qiskit qiskit-aer
python report/evidence/quantum_counting.py
```

The script deterministically seeds the Aer simulator (`seed=42`) and uses `shots=4096` per instance.

2.3 Implementation Notes

1. Grover operator G_F is built via `qiskit.circuit.library.GroverOperator` with a `DiagonalGate` oracle that puts a -1 phase on the marked computational basis states $\{0, \ldots, M-1\}$. The choice of *which* indices are marked is arbitrary for counting because the counting output depends only on $\lvert F^{-1}(1)\rvert$.

2. Controlled Grover powers on the precision register: for each precision qubit $j \in \{0, \ldots, t-1\}$ we apply the singly-controlled G_F gate 2^j times. This is a standard textbook realisation of QPE (see Nielsen & Chuang §5.2) and matches BHT'98's C_F up to gate ordering.

3. Inverse QFT on the precision register uses `QFT(t, inverse=True, do_swaps=True).to_gate()`. `do_swaps=True` makes the measured integer, interpreted with `prec[0]` as LSB, equal the estimated f (rather than its bit-reversal).

4. The Bhattacharya-style fold $f \leftarrow \min(f, P - f)$ is applied to collapse the two eigenvalue branches $e^{\pm 2i\theta}$.

5. First implementation attempt used a hand-rolled diffusion $H^n X^n \mathrm{MCZ}\, X^n H^n$ and produced systematically *wrong* $\tilde{t}$ (e.g. $\tilde{t} \approx 15$ for $t = 1$). Switching to Qiskit's canonical `GroverOperator` fixed this immediately. The bug was almost certainly a global-phase / diffusion sign convention (see §**??**).

2.4 Circuit Statistics

For $N = 16$, $P = 32$ the constructed circuit has 9 qubits (5 precision + 4 search), depth $\approx$ 1650–2180 after transpilation to Aer's basis, and $\approx$ 1650–2400 gates depending on t (fewer gates when the oracle is trivial, e.g. $t = 0$ never invoked here). Each 4096-shot run completes in well under a second on CPU.

3 Results vs. Paper

3.1 Headline Number: Theorem 5 Error Bound

The paper predicts $\Pr[|t - \tilde{t}| < \epsilon_5(P, t, N)] \geq 8/\pi^2 \approx 0.811$ where $\epsilon_5(P, t, N) = (2\pi/P)\sqrt{tN} + (\pi^2/P^2)N$. We measured this empirical probability at $N = 16$, $P \in \{16, 32\}$, $t \in \{1, 2, 4, 8\}$ using 4096 shots per configuration:

N	t	P	ϵ_5 (bound)	$\tilde{t}$ (best)	$\lvert t - \tilde{t}\rvert_{\text{best}}$	$\hat{P}$(success)
16	1	16	2.188	0.609	0.391	0.935
16	2	16	2.838	2.343	0.343	0.962
16	4	16	3.758	4.939	0.938	0.894
16	8	16	5.060	8.000	0.000	1.000
16	1	32	0.940	1.348	0.348	0.876
16	2	32	1.265	2.343	0.343	0.871
16	4	32	1.725	3.555	0.445	0.906
16	8	32	2.376	8.000	0.000	1.000

All 8 empirical success probabilities exceed the Theorem's $8/\pi^2$ floor. The maximum-probability bin $\tilde{t}$ rounds to the true t in every single row. Both the analytical error bound and the analytical success probability are reproduced by our real simulation.

Two sanity properties fall out for free: (i) doubling P from 16 to 32 approximately halves the additive error ϵ_5 (compare rows 1–4 vs. 5–8) — consistent with the $O(1/P)$ scaling of the dominant term; (ii) at $t = N/2 = 8$ the eigenvalue is $\pm i$ exactly ($\theta = \pi/4$, $f = P/4$), so the phase is representable exactly on both precision registers and we observe Pr(correct) = 1 and $|t - \tilde{t}| = 0$ identically.

3.2 Query-count / Quadratic-Speed-up Sanity

Classical exact counting of $t = |F^{-1}(1)|$ requires $\Omega(N)$ queries in the worst case; the paper's Corollary 2 gives $\tilde{t}$ accurate to a few standard deviations in $c\sqrt{N}$ queries. Our runs at $P = 16$ used exactly $P = 16$ oracle queries on $N = 16$, i.e. $4\sqrt{N}$ queries with $c = 4$. The observed additive error at $c = 4$ was ≤ 1 in 8/8 cases and the argmax bin always rounded to t. This is qualitatively consistent with the $c\sqrt{N}$ query budget of the corollary.

4 Failure Analysis (see also `failure_analysis.md`)

- **Diffusion-sign / eigenvalue-branch bug.** The first (hand-rolled) Grover operator gave systematically inverted counts ($\tilde{t} = 15$ for $t = 1$, $\tilde{t} = 12$ for $t = 4$, but $\tilde{t} = 8$ for $t = 8$ where the inversion is a fixed point). This looked like a sign error in the zero reflection S_0 or the missing overall (-1) in $Q = -(H^n S_0 H^n) S_F$. Root-cause fix: use Qiskit's `GroverOperator`. Lesson: never trust a from-scratch Grover diffusion without a QPE round-trip check.

- **Marker / Nougat unavailable.** Marker and Nougat are not installed on this host and no pre-parsed extraction for this arXiv id exists in the central corpus. We substituted a `pdftotext -layout` dump (see `extraction/`). For a 12-page text-native quantum-computing preprint the information content is unchanged; equations are rendered as ASCII, and every claim we needed was legible.
- **QFT deprecation warning.** Qiskit 2.5 emits a `DeprecationWarning` on `QFT(...).to_gate()` pointing at `QFTGate`; harmless for this run but a small forward-compat wart.

5 Verdict

REPLICATED. Theorem 5's error bound and success-probability guarantee are both borne out by real Qiskit statevector simulation on $N = 16$ across $t \in \{1, 2, 4, 8\}$ and $P \in \{16, 32\}$, and the maximum-probability estimator always identifies the correct integer count. The paper's main algorithmic claim reproduces cleanly with textbook QPE of `GroverOperator`.

Open Questions

See `report/open_questions.json` for the machine-readable version.

- **Q1.** How tight is the $8/\pi^2$ success-probability floor for small P and near-integer $f = P\theta/\pi$? Our smallest configuration ($t = 1, P = 16$: $f \approx 1.29$) already showed $\hat{P} \geq 0.93$, suggesting the floor is loose in the "comfortable" regime.
- **Q2.** What is the minimal P at which the max-probability rounded estimator $\lfloor \tilde{t} \rceil$ becomes unbiased for a given t/N? At $t = 1, P = 32$ we saw $\tilde{t} = 1.348$ (bias $+0.35$).
- **Q3.** Does the fold $f \leftarrow \min(f, P - f)$ silently double-count when f is exactly at $P/2$? Our $t = 8$ case has $f = P/4$ so this never triggers, but $t \approx N/2 - \delta$ configurations may show a discontinuity.
- **Q4.** Circuit depth grows roughly linearly in P (rows $P = 16 \rightarrow P = 32$ approximately doubled depth). For hardware execution on a NISQ device the depth 2000+ single-controlled-Grover chain is the bottleneck; can catalytic phase-estimation reduce it without loss of the Thm. 5 bound?
- **Q5.** We assumed marked $= \{0, \ldots, M-1\}$. Randomising the marked set should be invariant, but a numeric check would guard against oracle-implementation quirks in later Qiskit versions.